



Practice II: Building Coreference and Coauthorship Networks from Extracted WoS Records

Objective

In Practice 1, we extracted records from Web of Science.

In Practice 2, we use the exported WoS file to build two networks:

1. **Coreference network**
2. **Coauthorship network**

The input file for this exercise is: sir_ref.txt

1. Conceptual Overview

Coreference Network

In the coreference network:

- **Nodes** = publications from the extracted WoS records
- **Edges** = two publications share at least one cited reference
- **Edge weight** = number or proportion of shared references

Coauthorship Network

In the coauthorship network:

- **Nodes** = authors
- **Edges** = two authors appear on the same paper
- **Edge weight** = number of papers they coauthored together

2. Required WoS Fields

The script uses the following fields from the Web of Science plain-text file:

WoS field	Meaning	Used for
UT	WoS accession number	Publication node ID
AU	Authors	Coauthorship network
CR	Cited references	Coreference network
DI	DOI	Publication metadata
PY	Publication year	Later analysis
SO	Journal/source title	Later analysis

4. Expected Outputs

